

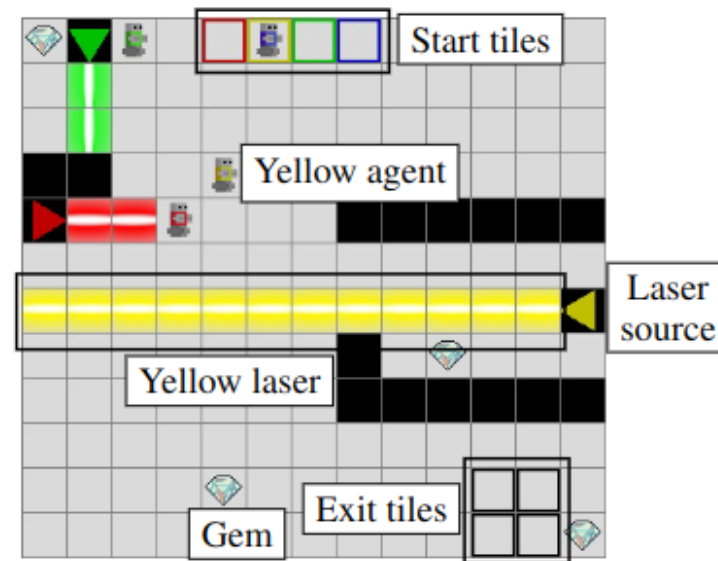


Procedural Generation of Solvable Cooperative Levels for the Laser Learning Environment

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Context & motivation

Laser Learning Environment (LLE)

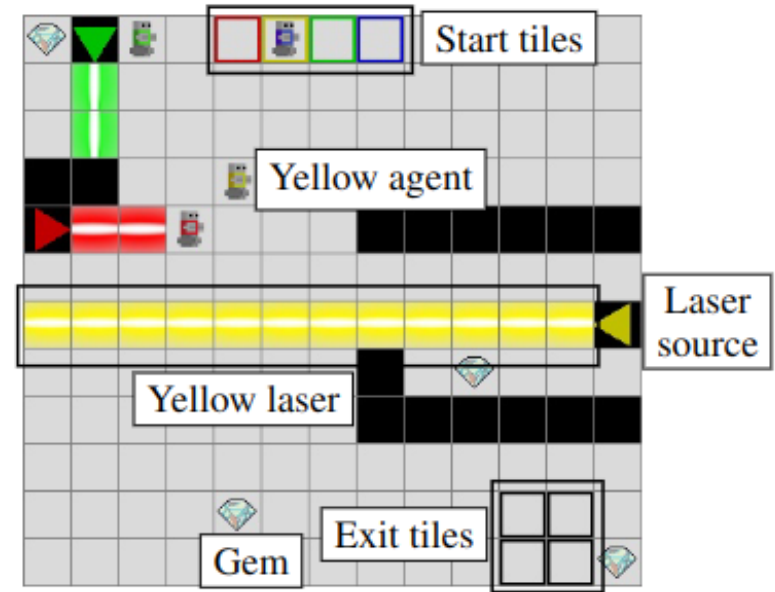
- 2D grid-based fully cooperative multi-agent puzzle
- Each agent must reach its exit tile **simultaneously** with all others
- Laser beams are passable only by an agent of the **matching color**
- Reward signal is sparse: zero credit for intermediate coordination steps

The LLE cooperative mechanic

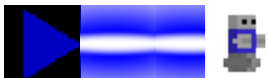
- A beam blocks **other** colours; each agent is **immune** to its own.
- Block your own beam → free a teammate's path (**locally unrewarded**).

Objective

Generate LLE levels **solvable** and that **require cooperation**.



LLE Level 6 — the hard target



Research questions

Generate levels **provably solvable** and **provably cooperative** — and use them to train MARL.

Formal (contributions)

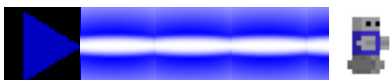
- RQ1 — verify **solvable**.
- RQ2 — verify **requires cooperation**.
- RQ3 — verification **inside a generator**.
- RQ4 — **control** the cooperation **structure**.

Empirical

- RQ5 — can MARL agents **learn** generated levels?
- RQ6 — does a **curriculum** reach Level 6?

Takeaway

Verification (RQ1–2) • Construction (RQ3–4) • Experiment (RQ5–6).



Bounded-horizon solvability $\rightarrow$ SAT

Model

- Level $L = (H, W, C, s_p, \mathcal{W}, \mathcal{S}, \mathcal{E})$, horizon $T_{\max}$.
- **Valid trajectory**: legal moves, no collisions, laser safety, all agents on exits at $T_{\max}$.
- **Solvable** = such a trajectory exists.

Variables (per step t)

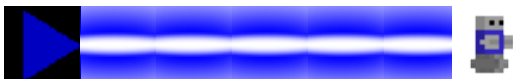
- $a_{c,x,y,t} \cdot b_{c,d,x,y,t} \cdot l_{c,x,y,t}$
- Constraints: **init** $\cdot$ **movement** $\cdot$ **lasers**

Verification tool

SAT solver decides whether $\Phi(L, T_{\max})$ is satisfiable
 $\Leftrightarrow$ a valid trajectory of length $T_{\max}$ exists.

Takeaway

Poly-size, poly-time encoding $\Rightarrow$ LLE solvability $\in$ NP and $\leq_p$ SAT.
NP-hardness **open**.



Detecting cooperation

Idea: keep the encoding, change **one** rule: beams no longer block same-colour agents, so the cooperation mechanic is **disabled**.

Verification criterion

L requires cooperation with $T_{\max} \iff \Phi(L, T_{\max})$ **SAT** and $\Phi_{\text{strict}}(L, T_{\max})$ **UNSAT**.

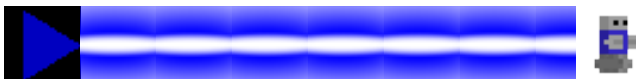
Solvable normally but impossible without blocking $\Rightarrow$ every solution uses the mechanic.

Two SAT calls:

1. Solver = UNSAT $\rightarrow$ UNSOLVABLE
2. Strict = UNSAT $\rightarrow$ COOPERATIVE
3. else $\rightarrow$ NON_COOPERATIVE

Horizon matters

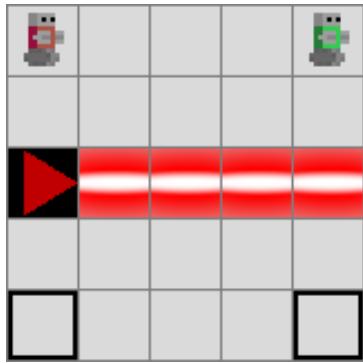
Label is relative to $T_{\max}$. a too-long horizon lets a detour bypass the beam.



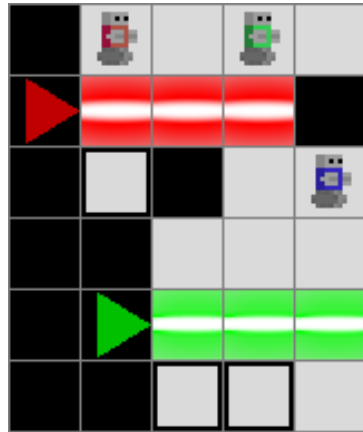
Cooperation profiles

From one SAT model → a **helper** → **beneficiary** graph

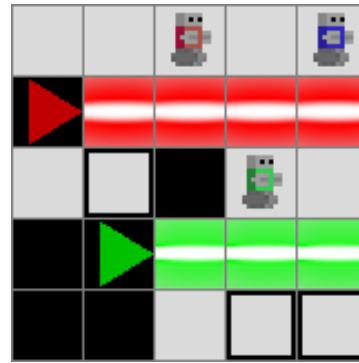
classify by priority: fully coupled $\succ$ mutual $\succ$ distributed $\succ$ chain $\succ$ asymmetric.



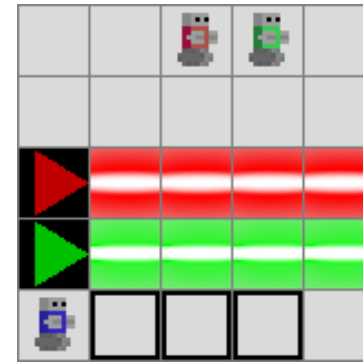
asymmetric



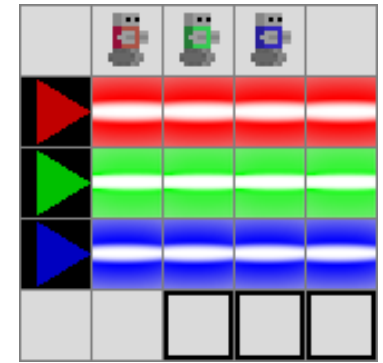
chain



distributed



mutual



fully coupled

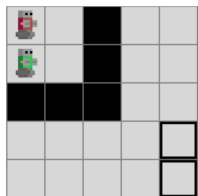
Takeaway

The profile is the **generation target** a generator can request.

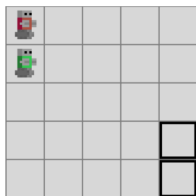


Generators

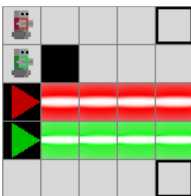
SAT as an acceptance oracle, in the loop. Solvability is the floor; **cooperation** and **profile** are optional layers.



(a) unsolvable
rejected



(b) solvable,
no coop.

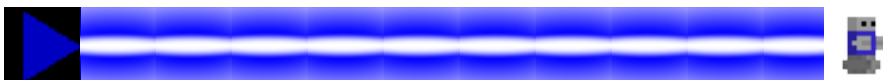


(c) cooperative
accepted

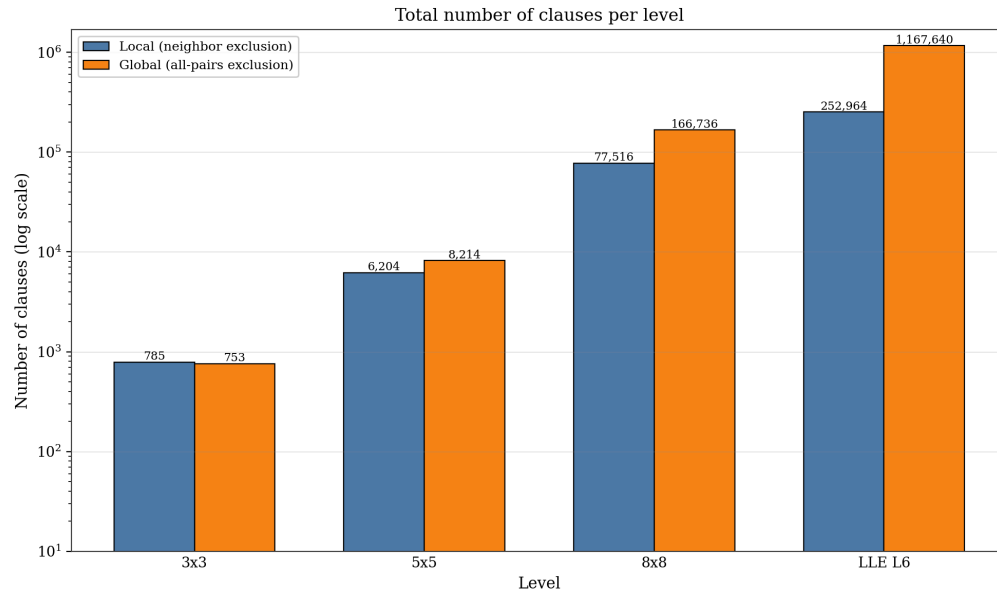
Generator	Profile tendency
Random	mostly asymmetric
Constructive	asym/mutual/distrib by laser count
Level-6-style	mutual by design

Guarantee

Every accepted level is **solvable** for a bounded horizon $T_{\max}$.



Result 1 – SAT encoding



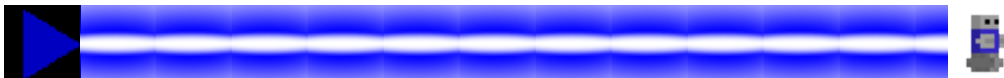
Enforcing “one position per agent per step”:

- **global** – pairwise over the grid, $O(p^2)$;
- **local** – neighbourhood-based, $O(p)$.

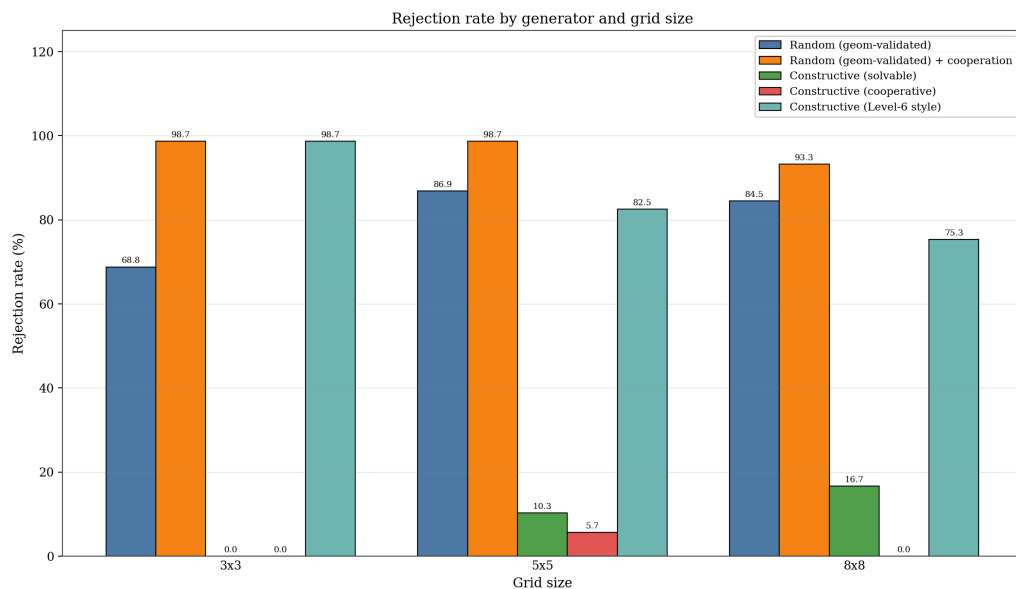
Local scales far better: **253k vs 1.17M** clauses on Level 6, and faster to solve.

Takeaway

The internal **encoding** drives cost. Local is the default.



Result 2 – Generator efficiency (RQ3)



- **Constructive (coop.):** $\approx 0-6\%$ — cooperation is **built in**.
- **Random (coop.):** up to 98.7% — coop. levels are a rare subset.
- **Level-6-style:** in between.

Takeaway

Construction beats blind sampling: near-free certified data.



Result 3 – Output diversity (RQ4)

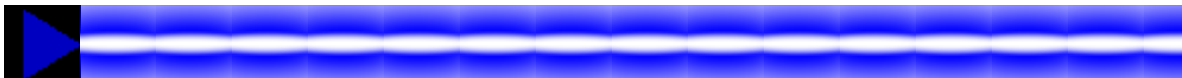


Profiles of accepted cooperative levels (8×8, 3 agents, 2 lasers):

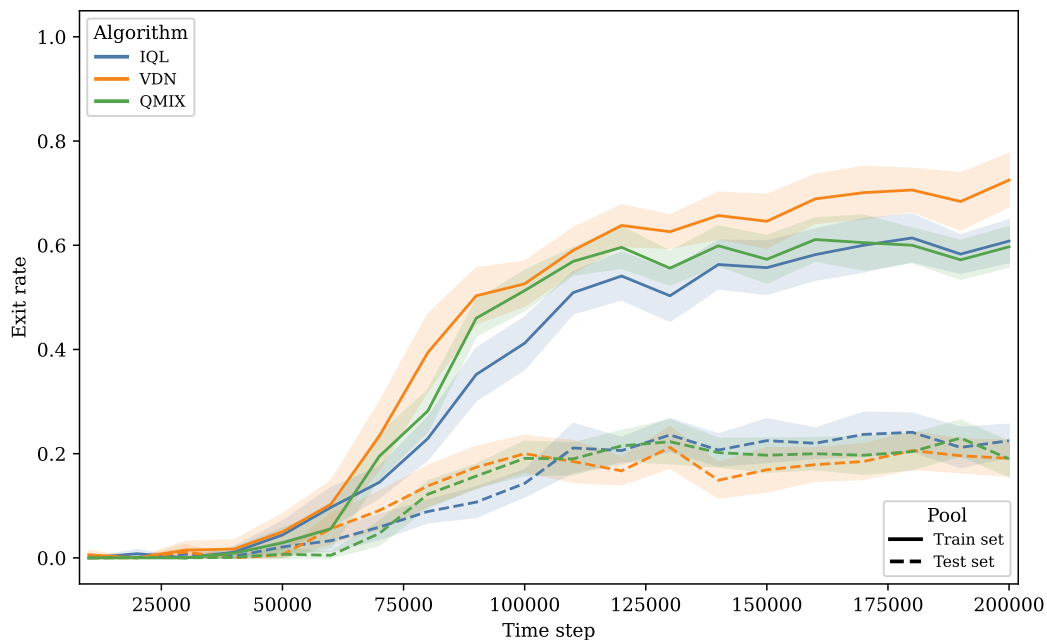
- **Constructive** → 92% mutual (by design).
- **Level-6-style** → 68% mutual (geometry).
- **Random** → mostly asymmetric.

Takeaway

We can **choose** the cooperation structure of generated levels.



Result 4 – Learnability (RQ5)

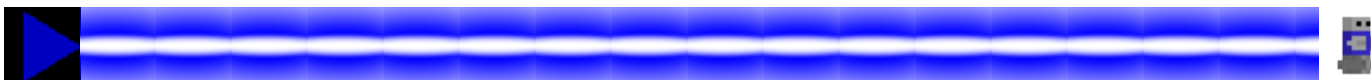


5×5, 2 agents, 1 laser · IQL/VDN/QMIX · 20 seeds.

- All reach non-trivial **training** success.
- Large **train-test gap** (≤ 0.52): they **memorise** the 20 levels.

Takeaway

Learnable at 5×5, but too few levels $\Rightarrow$ overfitting



Result 5 – Scaling data closes the gap (RQ5)

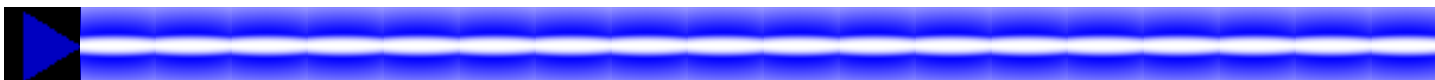


Same task but only **pool size** varies (20 $\rightarrow$ 100 $\rightarrow$ 500).

train	train	test	gap
20	0.65	0.14	0.50
100	0.47	0.28	0.19
500	0.43	0.43	0.00

Takeaway

Unlimited certified levels $\Rightarrow$ overfitting becomes generalisation.



Result 6 – Curriculum & its limit (RQ6)

Target	Grid	Coop.	Budget	Test
Learnability	5×5	asym	200k	0.20
Curric. order	6×6	asym	400k	0.17
Frontier	6×6	mutual	600k	0.00
Frontier	5×5	mutual	200k	0.00
Level-6	12×13	mutual	2M	0.00

- Reachable target: ordering doesn't beat **direct**; only data **diversity** helps.
- A **cliff** at **mutual** cooperation: success → **zero**, even at 10× budget.

Takeaway

Scoped **negative** result: the base **mutual** task is unlearnable by IQL/VDN/QMIX – not a curriculum failure.



Beyond the thesis: upstream

Solver, cooperation detector, and generators are **merged into the official LLE library** (laser-learning-environment[generator], PyPI, v2.9.0+).

Public API

```
lle.solve • lle.is_cooperative • lle.cooperation_level • lle.generate
```

Takeaway

Not just thesis artefacts — **usable tooling** for the LLE users.

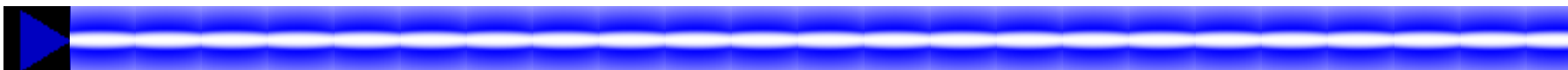


Conclusion

- RQ1 ✓ – SAT verification tool for solvability; in NP, $\leq_p$ SAT.
- RQ2 ✓ – strict counterfactual: coop. $\Leftrightarrow$ SAT $\wedge$ strict-UNSAT.
- RQ3 ✓ – SAT-as-oracle generators; every accepted level **certified**.
- RQ4 ✓ – five-profile taxonomy; generators steer toward mutual.
- RQ5 ✓ – certified levels learnable; scaling data closes the gap.
- RQ6 ✗ – value-based MARL can't learn **mutual** cooperation; no curriculum helps.

Takeaway

Central contribution: procedural generation coupled to **explicit certification**.



Future work

Theory & model

- Settle **NP-hardness** of LLE solvability.
- Add **gems** and true **void** tiles.
- Richer (time-indexed / weighted) cooperation graphs.

Learning

- **Make mutual coordination learnable:** dense / intrinsic rewards.
- Beyond value-based: actor-critics (MADDPG, COMA), MAVEN.
- Surface the *fully coupled* profile.

The certification framework is settled; the open question is **how far it extends**.



Thank you

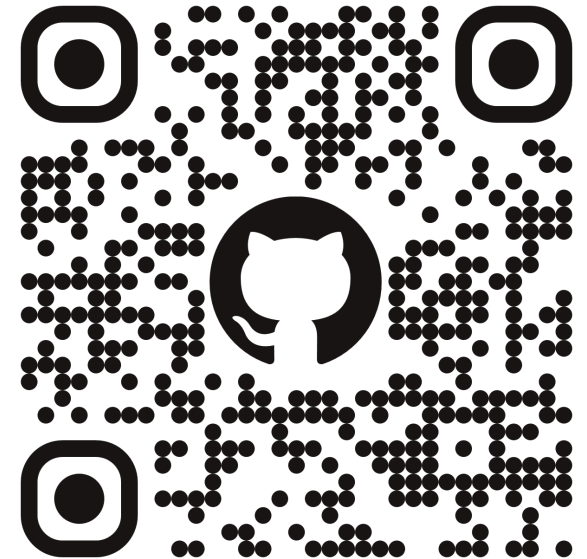
Contributions

1. SAT verification tool for LLE solvability.
2. Cooperation detector + five-profile taxonomy.
3. Solver-in-the-loop certified generators.
4. Merged into the official LLE library.

Key results

1. Local encoding scales; constructive generator $\approx$ free.
2. Scaling certified data closes the generalisation gap.
3. A learnability **cliff** at mutual cooperation.

Questions?



Backup slides

Backup – Full SAT encoding

Variables

- $a_{c,x,y,t}$: agent c at (x, y) at step t
- $b_{c,d,x,y,t}$: beam of colour c , direction d , active at (x, y, t)
- $l_{c,x,y,t}$: laser of colour c active at (x, y, t)

Constraints (4.1–4.13)

Init

- 4.1 agent start tiles
- 4.2 laser-source cells

Movement

- 4.3 forward consistency
- 4.4 global / 4.5 local uniqueness
- 4.6 backward consistency
- 4.7 no overlap / no following
- 4.8 victory • 4.9 stay-on-exit

Lasers

- 4.10 walls block beams
- 4.11 beam propagation
- 4.12 beam $\leftrightarrow$ laser link
- 4.13 no step on foreign laser

$O((n_a + s)p\tau)$ variables; clause count polynomial in n_a, p, s, τ .

Backup – Local vs global uniqueness

Global (A):

$$\bigwedge_c \bigwedge_t \bigwedge_{(x_1,y_1) \neq (x_2,y_2)} \neg a_{c,x_1,y_1,t} \vee \neg a_{c,x_2,y_2,t} \quad O(n_a \tau p^2)$$

Local (B): neighbourhood exclusivity over $\text{next}(x, y)$ + backward consistency

$$O(n_a \tau p)$$

Both admit the **same** satisfying assignments. Local is asymptotically smaller in p but pays a fixed overhead; below the **crossover** grid size the global form is more compact.

Level	3×3	5×5	8×8	Level 6
global	753	8 214	166 736	1 167 640
local	785	6 204	77 516	252 964

Clause counts. Level 6: 7× more clauses than 8×8 yet solves faster — search structure, not size.

Backup – Why the encoding is correct

Decision equivalence

For either movement formulation, $\Phi(L, T_{\max})$ is satisfiable $\Leftrightarrow$ a valid joint trajectory of length $T_{\max}$ exists.

SAT $\Rightarrow$ trajectory. Init fixes one start per agent; forward consistency + uniqueness (global) – or forward + local + backward (local) – force exactly one legal position per agent per step. Movement / collision / laser / exit clauses make the extracted trajectory valid.

Trajectory $\Rightarrow$ SAT. Given a valid trajectory, set a from positions and b, l from the deterministic beam dynamics; every clause family holds by construction.

Consequence. Poly-time, poly-size reduction $\Rightarrow$ bounded-horizon LLE solvability $\in$ NP and $\leq_p$ SAT. NP-hardness open.

Backup – Why the cooperation criterion holds

Verification criterion

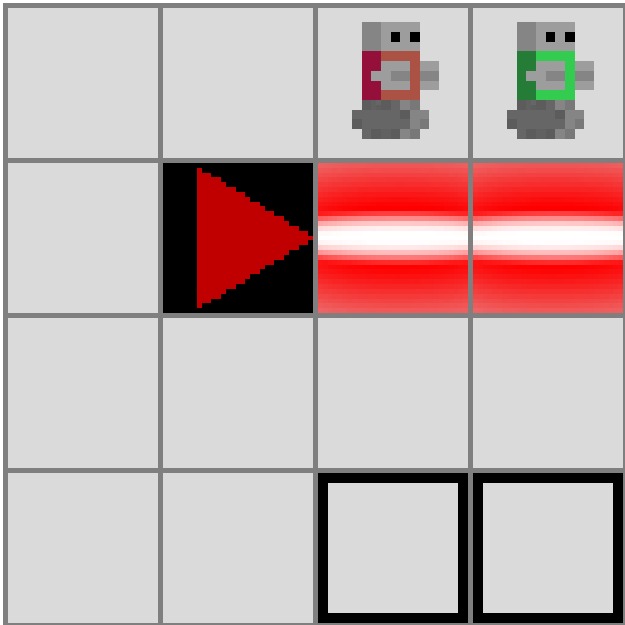
L requires cooperation at $T_{\max} \iff \Phi$ SAT and Φ_{strict} UNSAT.

($\Rightarrow$) Cooperation $\Rightarrow$ solvable, so Φ SAT. If Φ_{strict} were SAT, the strict trajectory would be a valid standard solution **without** blocking – contradiction.

($\Leftarrow$) Φ SAT $\Rightarrow$ solvable. If some standard solution used **no** blocking, both semantics give identical beams along it, so it would satisfy Φ_{strict} – contradicting UNSAT. Hence every solution blocks at least one beam.

Strict encoding changes one clause family: $b_{c,d,x',y',t} \leftrightarrow b_{c,d,x,y,t}$ instead of $b_{c,d,x',y',t} \leftrightarrow (b_{c,d,x,y,t} \wedge \neg a_{c,x',y',t})$.

Backup – Horizon-dependence of cooperation



Same level, three labels depending on $T_{\max}$:

Horizon	Solvable	Coop.
$T_{\max} \leq 2$	no	n/a
$3 \leq T_{\max} \leq 8$	yes	yes
$T_{\max} \geq 9$	yes	no

Too generous $\Rightarrow$ a **detour** fits $\Rightarrow$ under-detected. Too tight $\Rightarrow$ rejected as unsolvable. We pick the smallest horizon fitting a short solution + margin.

Backup – Selective-strict & necessary helpers

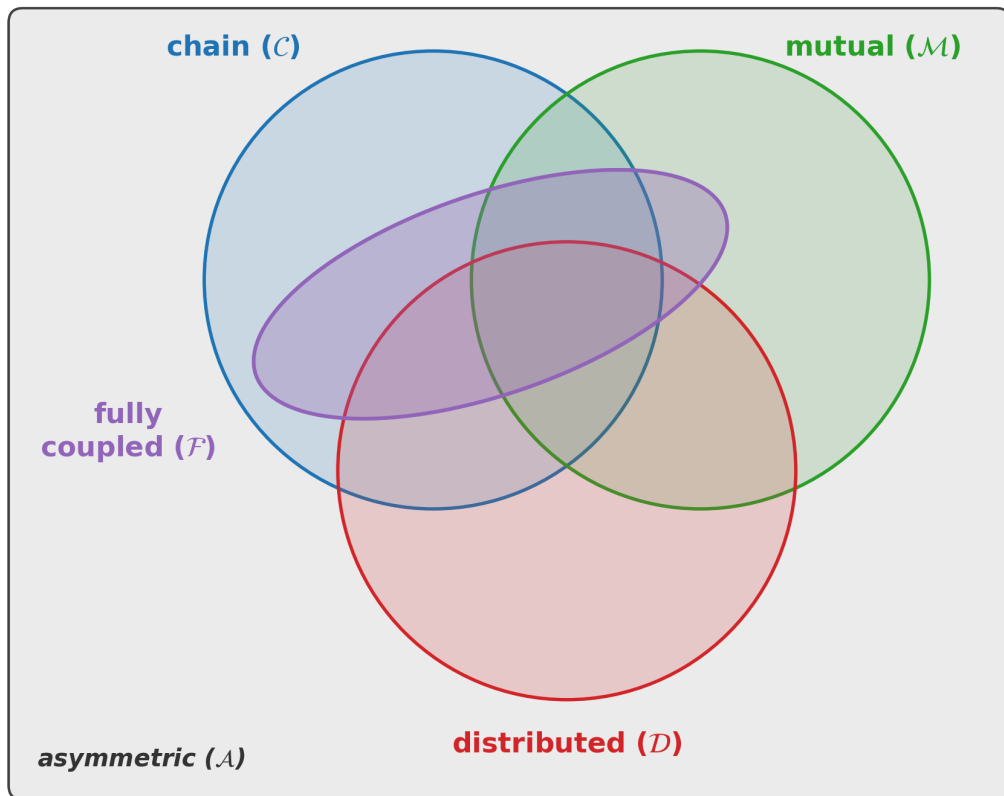
Selective-strict ($K \subseteq C_{\text{src}}$): strict clauses for colours in K , standard for the rest. $K = \emptyset \rightarrow$ standard; $K = C_{\text{src}} \rightarrow$ strict.

Necessary-helper set: for each colour c , run one selective-strict call with $K = \{c\}$; if UNSAT, c is **necessary**.

Invariants vs plan-dependent

- **Level invariants:** binary verdict, necessary-helper set.
- **Plan-dependent:** helper events, dependency graph, profile **label**.

Backup – Profile ordering structure



Predicates on G_L : $\mathcal{A}$ asymmetric (base), $\mathcal{C}$ chain, $\mathcal{D}$ distributed, $\mathcal{M}$ mutual, $\mathcal{F}$ fully coupled.

Only structural entailment: $\mathcal{F} \Rightarrow \mathcal{C}$. The rest are **incomparable** (pure chain / fan-in / reciprocal pair / 3-cycle witnesses).

Priority $\mathcal{F} \succ \mathcal{M} \succ \mathcal{D} \succ \mathcal{C} \succ \mathcal{A}$ is one linear extension, fixed by convention.

Backup – Constructive generator

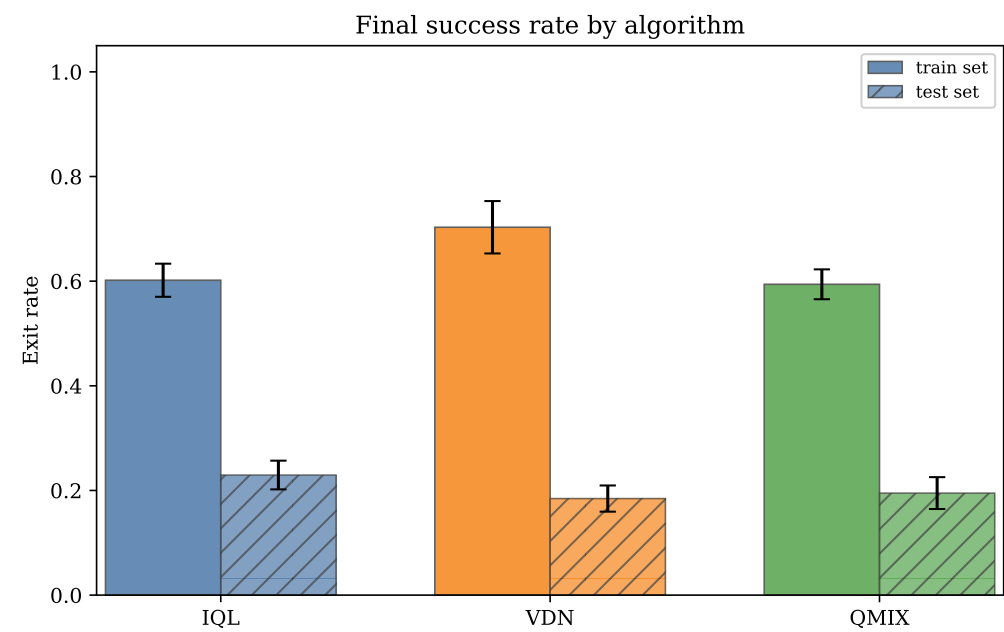
- Pick orientation; sample n_a distinct lanes; place start/exit at lane ends; reserve lane cells.
- Cooperative mode plants a dependency per laser: distinct colour ($n_l \leq n_a$), beam crossing the lane band; reserve the full beam segment before placing walls.
- Shuffle remaining cells; place n_w walls; verify with the SAT oracle.

Profile by laser count

$n_l = 1 \rightarrow \text{asymmetric} \cdot n_l = 2 \rightarrow \text{mutual} \cdot n_l \geq 3 \rightarrow \text{shifts to distributed.}$

Level-6-style: clustered starts/exits + a laser corridor $\Rightarrow$ mutual like Level 6.

Backup – Learnability detail (5×5)



Algo	Train	Test	Gap
IQL	0.60 ± 0.03	0.23 ± 0.03	0.37
VDN	0.70 ± 0.05	0.18 ± 0.03	0.52
QMIX	0.59 ± 0.03	0.20 ± 0.03	0.39

20 seeds, 200k steps. VDN fits train best yet generalises worst – capacity spent memorising.

Backup – Curriculum ordering (reachable 6×6)

Four budget-matched schedules, 400k steps, 6×6 / 2-agent / 1-laser (asymmetric):

Condition	Train	Test	Gap
Mixed (dom. rand.)	0.39	0.22	0.17
Direct	0.41	0.17	0.24
Forward (easy→hard)	0.39	0.15	0.24
Reverse (hard→easy)	0.10	0.07	0.03

- **Forward** curriculum does **not** beat **direct** (budget stolen from the target).
- **Reverse** collapses → **catastrophic forgetting**.
- Only **mixed** helps — that is data **diversity**, not ordering.

Backup — Why mutual cooperation is unlearnable here

1. **No reward to amplify.** Mutual reward arrives only after the whole jointly-gated sequence; return stays negative. A curriculum amplifies a faint signal — it cannot create one.
2. **No partial skill in easier stages.** Asymmetric coop. admits partial credit; mutual is a deadlock-style discontinuity — staging laser count ($0 \rightarrow 1 \rightarrow 2$) ended at ≈ 0 .
3. **Representation limit.** VDN/QMIX monotonic factorisation enforces IGM; the mutual move is individually suboptimal for **both** agents (relative overgeneralisation). IQL has no joint value.

Takeaway

Nonzero **train** but zero **held-out** = memorised trajectories, not a policy.

Backup — Related work & positioning

- **Cooperative MARL / LLE** (Moltinghen et al.): coordination bottlenecks; VDN, QMIX, MADDPG, COMA, MAVEN.
- **Dependency structure**: coordination graphs (assume structure) vs ours (recovered from a SAT certificate).
- **PCG under constraints**: search-based / ASP oracles; PCGML (no guarantees); PCGRL (opposite direction).
- **Curriculum / env. design**: POET, PAIRED (adaptive, adversarial) — ours is a **static** certified generator.
- **Compilation**: SAT-planning (SATPLAN), compilation-based MAPF (we add laser propagation + strict counterfactual).
- **Formal methods + RL**: shielded RL filters **actions**; we filter **training material**.

Backup — Scope & threats to validity

Scope. Exit-reaching task only; gems and scoring out of scope; void tiles modelled as walls (valid only when no beam crosses a void). $1 \leq n_a \leq 4$ is an engine bound, not an encoding limit.

RQ6 caveats (we do not overclaim):

- Narrow algorithm family (IQL/VDN/QMIX); centralised critics / coordinated exploration untested.
- Sparse, unshaped reward; a dense signal could change the bottleneck.
- Small seed counts on mutual targets — enough to establish **zero**, not small effects.

Cooperation criterion is horizon-relative; the profile **label** is plan-dependent.